Critical Decision Making in Emergency Nursing: Findings from a Qualitative Study

Sudeep Hegde, PhD[a], Jessica Arora[b], Rebecca L. Berg[c], Tracy C. Kim[d], Daniel Hoffman[e], Natalie Benda, PhD[f], Emilie Roth, PhD[g], Shawna Perry, PhD, MD[h], Rollin J. Fairbanks, MD, MS[i,j], Ella Franklin, RN EDAC[i,j], A. Zachary Hettinger, MD, MS, FACEP, FAMIA[i,j], and Ann M. Bisantz, PhD[k]

[a]*Department of Industrial Engineering, Clemson University,*
[b]*Department of Systems and Information Engineering, University of Virginia*
[c]*Mozarc Medical*
[d]*SHRM*
[e]*Mission Technologies, a division of HII*
[f]*School of Nursing, Columbia University*
[g]*Roth Cognitive Engineering*
[h]*Department of Emergency Medicine, University of Florida*
[i]*Medstar Health Quality & Safety, MedStar Health*
[j]*Department of Emergency Medicine, Georgetown University School of Medicine*
[k]*Department of Industrial and Systems Engineering, SUNY University at Buffalo*

First Author:
Sudeep Hegde, PhD
Sudeeph@clemson.edu
Department of Industrial Engineering
272 Freeman Hall, Clemson University
Clemson, SC 29634-0920

ABSTRACT

While decision making has long been studied in various nursing domains, there has been scarce focus on emergency nursing beyond the triage process. This research demonstrated the novel use of Critical Decision Method to better understand decision making of nurses during the ongoing care of patients in the emergency department. Ten nurses from emergency departments across a large healthcare system were interviewed for descriptions of incidents from their practice involving one or more critical decisions. The qualitative analysis of data provided important insights into decision making, including the identification of important decision categories as well as the role of experience in emergency nursing. Further, the study identified nurses' perceptions of novices and the role of experience in decision making. The research also identified the role of nursing's clinical reasoning, critical decision making and interventions as essential to the effective care for patients in the dynamic environment of the Emergency Department.

KEYWORDS

Critical Decision Making, CDM, Emergency Nursing, Human Factors

INTRODUCTION

Nurses' decision making has long been a subject of study. Decision making in nursing practice is a complex process and considered a critical part of the nursing profession (Johansen & O'Brien, 2016). Research has focused on various healthcare settings including critical care, and conditions such as sepsis, cardiac arrest (Aitken et al., 2009; K. A. Hoffman et al., 2009). Studies pertaining specifically to nurse decision-making in emergency medicine have worked to better understand nurse decision-making during the triage process, an initial step used to determine how sick the patient is and what level of care they require (Chung, 2005; Cioffi, 1998; Considine et al., 2007; R. R. Hoffman & Coffey, 2004; Reay et al., 2020; Agius et al., 2025). While nurses do have responsibility during triage, this process only accounts for a small part of emergency medicine care. Nurses perform a variety of other, essential tasks including: taking patient history, counseling, administering medications, facilitating diagnostic tests, determining patient illness severity while simultaniously responding to health system demands (Hobgood et al., 2005; Potter et al., 2004; Jackson et al, 2021). Therefore, critical decision making could be demonstrated during the entire patient care process. Other research on decision making in emergency medicine settings has focused on physicians (Schubert et al., 2013), or on specific clinical procedures such as catheterization (Mizerek & Wolf, 2015). These and other studies have demonstrated the difficulty and complexity associated with decision-making in emergency medicine due to time pressure, high mental workload, and frequent interruptions (Croskerry, 2005; Laxmisan et al., 2007). Recent studies have started to look into implementing artificial intelligence tools into the emergency department to aid in decision making and interpreting labs (Ten Berg et al., 2023; Huang et al., 2023). To the authors' knowledge, there is limited research focusing on critical decision making more broadly in emergency nursing.

Many of the studies on nurses' decision making have explored differences between novice and expert nurses, dating back to the seminal work of Benner, who characterized multiple levels of nurse expertise based on the Dreyfus Skill Acquisition Model, ranging from novice, through intermediate stages, to expert (Benner, 1984). Experienced nurses are able to make decisions more quickly using intuition, case-matching with previous experiences, and internalized knowledge of decision protocols (Cioffi, 1998; Gerdtz & Bucknall, 1999). However, beyond efficiency, there is no conclusive evidence on the influence of experience on decision making. In fact, Considine et al (Considine et al., 2007) found that experience, when considered merely as number of years in a role, does not significantly relate to decision making. While these studies identify differences in DM between specific 'stages' of experience, it is intrinsic to all decision making (Cioffi, 2001). Therefore, there is a potential to understand the role of experience, regardless of level, in the decision making of emergency nurses.

A number of qualitative techniques have been used to study nurse decision making. The Critical Decision Method (CDM), in particular, has been utilized to elicit specific strategies used by experienced nurses in specialty areas including acute care, pediatrics, and the neonatal intensive care unit (Crandall & Getchell-Reiter, 1993; Ebright et al., 2003; C. Patterson et al., 2016). The current research employs the CDM to study the types of decisions made by emergency nurses in critical situations. The study will also identify the role of experience in the decision-making of emergency nurses. Implications for using these insights to design training for emergency medicine clinicians and develop enhanced decision-support for the work domain will be discussed.

METHODS

Protection of Human Subjects

Approval was obtained from the MedStar Health Research Insitute Institutional Review Board prior to the study.

Participants

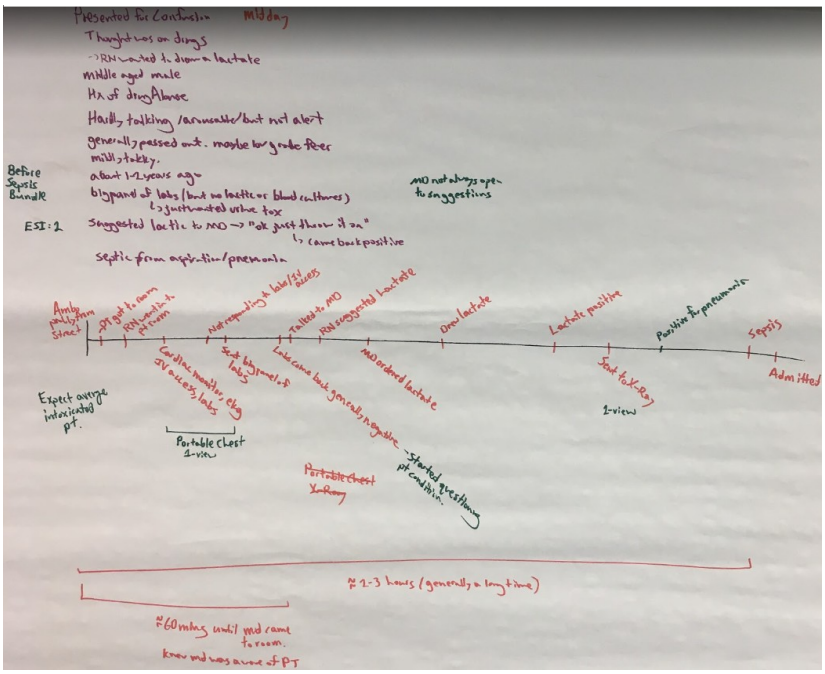
Participants were recruited from a large, academic healthcare system that sees in the Mid-Atlantic United States through flyers, email, and word of mouth. The health system sees over 1,000,000 outpatient visits per year and the emergency departments have approximately 500,000 patient visits per year. Data was collected on demographic information of the participant, including current ED experience and total ED experience (in years), role, age, gender, race and ethnicity. Participants were compensated for their time.

Data Collection

CDM interviews took place in a private conference room and lasted between 60 and 90 minutes. A voice recorder, electronic notes, and a flipchart were utilized to document information. CDM is an interview method that supports cognitive task analysis, was used to collect and analyze data (Hoffman et al., 1998). The purpose of the CDM is to elicit domain knowledge from experienced practitioners, in the context of situations characterized by time pressure, high information content and variability (Klein et al., 1989). CDM includes four interview "sweeps": selection of an incident, construction of a timeline, deepening, and "what if" queries. During the incident selection sweep, participants were asked to remember an incident with a specific patient where their decision making had a direct effect on the outcome. After an appropriate incident was selected, participants were asked to go through in detail the chronological order in which events happened in the incident. This timeline was documented during the sweep, on a flipchart (see Figure 1 as an example), to allow the participant and

researchers to review the critical points where there may have been a shift in understanding or where decisions and actions were made. The third sweep asked questions to obtain additional information including cues, goals, options, and experience. At the end of the interview, the participants were asked "what if" queries such as hypothetical changes in the incident, what aids would have been useful, and how less experienced nurses would have approached the incident. In addition, the participants were asked to compare, hypothetically, their decision making in that situation with that of a novice nurse in their place. The term 'novice' was not explicitly defined.

Figure 1: Timeline of events for a scenario developed on a flipchart during the interview



The interview guide was adapted from Patterson et al. (2016).

Data Analysis

The electronic notes from the interviews were edited with information from the audio recordings. An adaptation of the Structured Approach by Wong (2004) was used to analyze the data (Harenčárová, 2015). The adapted approach in this study included the following steps:

- *Decision charts* were created showing the decision process on a timeline and detailing information such as decisions and actions.
- *Incident summaries* were formed from the interview notes as well as the audio recordings. These focused on a wide view of the incident in a story-like format. Information included details about the participant as well as other actors and the environment in which the incident happened.
- *Decision Analysis Tables (DATs)* were created to identify various aspects of the decision making for each incident, including cues, situation assessment, decision or action, and goal of the decision. DATs are based on the Decision Requirements Tables reported in Hoffman (2005). Each decision was categorized according to the type of decision taken, such as diagnosis, intervention, monitoring, communication and prioritization. Decisions and actions were labeled as such only if they were taken by the nurse-participant themselves. Decisions and actions taken by coworkers, such as physicians or other nurses, were considered as cues – essentially, input to the nurse-participant's decision-making process. In addition, the DAT also described potential implications of the nurse's experience to the decision taken.
- *Collation and comparison of common items of interest:* The analysis was aimed at identifying the role of experience in nurses' decision making. Accordingly, as a first step, the DATs were reviewed to identify themes that characterize decision making. A separate category was created consisting of themes related to the implication of experience for each case. In addition, all participants' responses to the question "What would a novice nurse have done in this situation?" were considered as a separate category to identify decision-making themes. Themes were initially identified separately by two analysts (SH and JA), who then reviewed each other's themes across all interviews. Similar or overlapping themes were consolidated, and differences in themes were reconciled between the two analysts and finalized.

RESULTS

Participants

Ten emergency nurses were recruited in the study. The experience of the nurses in emergency care ranged from less than 2 years to 34 years. Four nurses had about 1.5 years of experience, three had between 4 and 7 years, two between 8 and 10 years, and one was an ED nurse for 34 years. Five of the participants had additional years of experience (1-8.5 years) in the ED in a different role, such as technician. The specific nursing roles included staff nurses (registered nurse) resource leader. Nurses' ages ranged from 24 to 58 years, generally proportionate with experience.

Incident Characteristics

During each interview, the participants were asked to describe incidents or 'scenarios' from their practice, which involved critical decision making. The participants described 2-3 scenarios each, and a total of 21 scenarios were described. Scenarios involved a patient arriving at the ED by themselves, with family members, or brought in by ambulance. An initial assessment by the nurse is followed by appropriate steps, such as obtaining blood and urine samples and interventions like placing intravenous lines and placing a patient on a monitor. The scenarios include communication between co-workers in the ED, including the physician, and staff from other areas of the hospital if necessary. Each incident usually included multiple decisions across the progression of events. Table 1 summarizes the key characteristics of the scenarios discussed in the interviews.

Table 1: Key characteristics of the scenarios

Interview #	Incident #	Patient gender	Patient age	Chief complaint	General Incident info
1	1	Male	older (75-85)	Mild arm pain	Patient (Pt) who looked healthy, went under cardiac arrest very suddenly; had to make a decision because the nurse was seeing another patient at the time, but this was "her" patient
1	2	Female	80	Shortness of breath	Pt did not want to be intubated unless it was extreme emergency, doctor wanted to intubate, did not intubate because respiratory therapist did not see it to be necessary
2	3	Male	50s	Difficulty breathing	Pt had a blood pressure (bp) of 244, RN was present in room for most of the patient's stay
2	4	Female	N/A	Debilitating abdominal pain radiating to her back with high fever, elevated HR, was diabetic and looked pale	Nurse got labs and medications started before the doctor came in, novice would have waited for doctor to order sepsis medications
3	5	Female	mid 40s	Unresponsive	Pt was vomiting blood. RN noticed brown tinge around pt mouth, suspected massive internal GI bleed. CAT scan showed ruptured pancreas. IVs placed and blood was given. RN saw improvement from vital signs, but not in patient appearance. Immediate surgery was ordered.
3	6	Male	19	Vague complaints	Pt had a slight fever and a barely elevated heart rate (HR), when the RN reassessed him he had a higher temp and HR, had to get doctor after this
4	7	Male	60		Pt had a history of stroke, he could not move anything
4	8	Male	9	Chest pain	Pt had been on amphetamine/ dextroamphetamine for the last year and needed to be transferred to cardio
4	9	Female	55	belly pain and shortness of breath	Pt come in with history of asthma, obtained EKG, gave morphine
5	10	Female	50s-60s	headache, confusion	Pt's presentation was somewhat different than a regular migraine. RN coordinated additional tests (neuro, ABG) to prove this wasn't a 'normal headache'
5	11	Male	37	chest pain/pressure	Patient was tachypneic, stabbed and shot in the head
6	12	Male		active hit on him	Pt had an infection in the foot from a gunshot wound; had to take security precautions
6	13	Female	40s	asthma	Pt. with suicidal ideation came in, but RN decided to stay with asthma patient

7	14	Female	elderly	Weakness and fatigued	Elderly woman came in very tiny, GI bleed, sitting in pool of blood,
7	15	Male	middle-aged	Confusion	Drug abuse, septic
8	16	Male	57-58	throwing up blood	History of bowel surgery a few years ago, was not too serious when first came in, got worse in couple hours, went to ICU
8	17	Female	early 70s	trouble breathing	Pt came in from assisted living, wasn't able to verbalize very well, dr decided to intubate,
9	18	Female	40s	chest pain, shortness of breath	Pt in SVT, dr unable to get into the room, monitored vitals, dr told nurse that she knows what to do and gave her the OK to do it
9	19	Female	60s	abdominal pain, nausea, and vomiting	Made sure RN knew this was the most critical pt, doc wanted sepsis work up, pt needed to go down to CAT scan, moaning, dry heaving
10	20	Male	upper 50s	Shortness of breath and trouble breathing at home	RN made the decision to put pt on BiPAP
10	21	Female	13	Heart Hurt	Made decision to do EKG on pediatrics patient and decided not to let the pt move after what RN saw

Decision Analysis Tables

The Decision Analysis Tables (DATs) recorded themes related to the nurses' decision making, organized under various categories. For instance the excerpt from a DAT shown in Table 2 relates to an incident involving a middle-aged female patient who was found unresponsive at her house and was brought to the ED by emergency medical service (EMS). The paramedic provided an initial cue that the patient might have suffered an overdose of an unknown substance and that she was not responding to naloxone (opioid antidote). The RN in this case proceeded to examine the patient and identified a number of cues (e.g. brown tinge around the mouth) that she noticed. Her initial assessment based on these cues was that the patient has had a massive internal GI bleed.

Table 2: Decision Analysis Table (DAT) for an examplar scenario

Cues	Situation Assessment	Action/Decision	Why	What for/Goals	Category of Decision	Experience Implication
Middle-aged woman was found unresponsive at her house						
Medic kept saying they thought it was an overdose but she was not responding to narcan		RN assessed patient herself				Some nurses may have gone by what the medic said and not have assessed the patient themselves
Pt had unstable vital signs: low BP, was tacky (increased HR)						Newer nurses don't speak up as much – believe docs always right
RN noticed brown tinge around her mouth, stains on teet, and that she was flaccid	Thought it was a massive internal GI bleed	Did stat CAT scan of abdomen (decided by doctor) right after the paitent was evaluated	Tinge around mouth – indicates GI bleed	Ascertain cause of bleeding	Diagnosis	Knew this (tinge) from previous experience with GI bleeds
Doctor decided to do CAT scan. Surgery team didn't see anything GI related from the dry reading of the CAT scan	Had pancreatic rupture (discouvered through CAT scan) and needed surgery	The surgery team came down, looked at it				Didn't know enough about pancreatic rupture and symptoms
	Was night time – surger material were not "prepped" so staff neded to make a quick decision	Surgery required	Treat pancreatic rupture and internal bleeding		Intervention	Junior and "lot of other" nurses would have done what doctor said – task driven
"H&H" [hemoglobin & hematocrit] vitals were dropping	Patient needed blood urgently	Supply blood to patient: RN got 4 units of uncross	Patient losing lot of blood due to internal bleeding		Intervention	New nurse would have felt alone

		matched blood				
		"We were standing on chairs squeezing blood into this lady"	To make up for the blood pt had lost for about 1-1.5 hours until she was able to go to the OR	Sustain patient while awaiting surgery		Previous experience of rushing blood into patient before surgery helped New nurse would have felt uncomfortable doing this

Decision types and characteristics

Within the DATs, various types of decisions were identified within each incident described by the nurses. The most commonly identified decision categories were:

- *Problem Identification:* Decisions in this category related to assessing and ascertaining the patient's condition based on one or a combination of cues, including the presenting symptoms and changes in patient condition over time, test results, and information from paramedics. For instance, a nurse noticed a brown tinge around a patient's mouth, stains on her teeth, and that she was flaccid. Based on these cues, the nurse suspected a massive internal GI bleed. In response to this clinical assessment, and to evaluate for other pathology, the nurse coordinated a CT scan of the patient's abdomen. This category also frequently involved initiating appropriate labs and reviewing the results as these were received in coordination with the rest of the clinical team.
- *Intervention:* Decisions taken based on the patient presentation and working diagnosis in order to act on the patient in some way, to alleviate or prevent a condition, were considered as interventions. For example, for a patient who was having breathing difficulty and becoming hypoxic, the nurse proactively had the respiratory therapist initiate additional breathing support with bilevel positive airway pressure (BiPAP) on the patient. Other actions coded as interventions included administering medications, or initiating cardiopulmonary resuscitation (CPR) on a patient who no longer had a pulse they could feel.
- *Monitoring:* This category includes decisions which involve looking for cues or observing trends and patterns in the patient's presentation, vitals, and other symptoms with the objective of detecting a change that might require an intervention. Further, in cases where a patient's condition was fragile and unstable, the nurse would continue to proactively monitor the patient more frequently. For instance, for a patient whose EKG showed an arrhythmia, the nurse decided to leave the EKG leads in place to facilitate a repeat of the EKG while monitoring the patient's response to medication: *"… never even unhooked EKG to keep an eye on his heart rate…(I) needed to see if medication was working"*.
- *Prioritization:* Nurses prioritize tasks for each patient and between multiple patients based on a number of factors including clinical needs or urgency and availability of resources. For instance, one nurse prioritized taking vitals and bloodwork over doing paperwork, which was less time critical and could be done later. In another example, a nurse was caring for two serious patients; one with suicidal ideation and one who had been suffering with asthma for four days. She prioritized staying with her asthma patient because she was not comfortable with the patient's degree of breathing difficulty.
- *Communication/Coordination:* This category describes decisions to communicate and coordinate information about plans and actions with coworkers, and patients and their families. These included updating the doctor about a patient's status, asking a patient about their symptoms, updating families about what is happening to the patient, or discussing next steps with nurses. The interviews also showed how nurses coordinate among themselves in order to accomplish various tasks. For example, while one RN administered an intravenous medication, her colleague gave the required saline flush immediately t after to achieve the necessary rate of medication administration.
- *Preparedness:* This category refers to actions taken, or decisions made to prepare for a potential change in the patient's condition that may require an urgent response. For example, a nurse brought the defibrillator on her way back to the room and had a second dose of medication in case either were necessary. Nurses commonly start an intravenous (IV) to expedite the administration of medications once ordered by providers. Similarly, nurses who have developed trusting relationships and established patterns of practice with specific providers are able to anticipate the sequencing of interventions for particular types of patients and thus make decisions accordingly. In general, the ability to anticipate potential changes or requirements for a specific clinical condition is tied to experience – nurses who are familiar with the patterns in the work environment, are able to anticipate the potential needs of the patient and team.

Role of Experience

Experience in the ED was found to play an important role in various aspects of decision making across all participants. The DAT recorded the role of experience in the interviews ("experience implication") based on participants' statements that were indicative of experience underpinning a decision, such as a nurse knowing what signals to heed or to ignore as false alarms. Themes reflecting the role of experience are described below:

Clinical Judgment and Problem Identification (4 of 20 incidents)

The role of experience was reflected in the use of cues by nurses to inform their assessment of the situation and make clinical judgements. A variety of cues were described by nurses, a majority of which were clinical in nature, such as relating to the patient's presentation, their medical history, medical alarms and alerts, and lab reports. Regarding alerts, for instance, a nurse summarized the role of her experience, as *"the longer I've been here, the more I've learned not to just ignore it [a coding alert], but to look at the monitor and see which room it was coming from."* Cues also related to information provided by other individuals, such as the patients themselves or their families, paramedics who bring the patient into the ED, and coworkers such as physicians and nurse colleagues. For instance, a diabetic patient was diagnosed with kidney issues and urinary tract infection, had high fever, elevated heart rate, and had pain on right side. Additionally, the nurse noticed that the patient looked pale, but was alert and oriented. Her knowledge of the sepsis protocol and experience with patients with sepsis led the nurse to consider the pain on the patient's right side as an important cue pointing towards a sepsis and possible kidney infection.

While cues were used for assessment in all the scenarios described by the nurses, this theme relates specifically to the use of cues to verify or ascertain a previous clinical judgement. In cases where cues were ambiguous, additional assessments and conversations with patients and families were required to clarify the situation. For instance, a nurse found that a patient who presented with migraine symptoms, but whose vitals were stable and normal, was not very willing to engage in conversation or answer questions and was sensitive to light and sound. Although a migraine protocol was initiated, the "*cues didn't quite add up* [for a migraine diagnosis]". It also didn't seem like a stroke, as in such patients, *"a weird yawn is usually seen, but not in this case"*, and this patient's vitals were normal. Drug-induced sleepiness was ruled out because such patients "*usually have a lean and then pop back up"*, which wasn't observed in this case. The combination of cues led the nurse to suspect a neurological issue as an alternate hypothesis. Therefore, she arranged for a CT scan, in consultation with the physician, to evaluate a potential neurological condition.

Nurse Intervention (10 out of 20 incidents)

Intervention refers to the willingness of the nurse to take appropriate action in response to a critical or life-threatening patient condition. The actions may include established protocols or steps taken in in the context of the practice culture and collaborative relationships. In other words, nurses intervened when they saw critical (potentially life-threatening) deterioration to prevent further deterioration, even if it often meant taking action in advance of a provider assessment. The role of experience in driving nurses' intervention could be summed up in the following quote from one of the participants: *"When you're trained, it's very systematic in that you have to do things in this order or this way.* (Experience teaches you) *to be flexible, may be do things in a different order"*. Nurse intervention was seen in all the categories of decisions. As an example, in terms of problem identification, a nurse with 1.5 years of experience as nurse, but with 10 years in the ED in other roles, was concerned about the way a patient was presenting. They had a 'weird' stare and their mother reported that the patient had a history of stroke. Recognizing the possibility of the patient having suffered a stroke, the nurse initiated the stroke protocol to order a CT scan without waiting for the attending, a practice that is supported by standing nursing protocols that recognize the critical time sensitive nature of strokes and the potential delays waiting for an electronic order can take. Examples of nurse intervention also included the nurse advocating for the patient, even if it meant disagreeing with the doctor. For instance, for a patient who had expressed a strong preference to not be intubated, the nurse decided to express her concern regarding the doctor's order for intubation ('intervention' decision). She perceived that the patient would be able to breathe on their own and advocated for the patient's wish to not be intubated. She also used a similar opinion from the respiratory therapist to support her assessment.

Knowledge (6 out of 20 incidents)

Participants highlighted their knowledge of typical therapeutic plans for specific patient conditions. Some of this knowledge is a direct result of nurses' formal education, such as knowing the indications for a specific drug and some is gained during standardized certification programs such as Advanced Cardiac Life Support (ACLS). More frequently, however, knowledge was described as resulting from experience in the context of clinical practice. For instance, a nurse's recognition of supraventricular tachycardia on EKG and her knowledge of the risk of patient deterioration associated with the rhythm informed her decision to interrupt the attending physician, who was attending to a critical patient, to make them aware of her patient's condition and to confirm her interpretation of the EKG. The physician confirmed the nurse's interpretation and agreed that the nurse should implement the appropriate treatment for the patient as prescribed by ACLS protocol. Knowledge and proficiency based on experience were found to underpin clinical judgment and supported all categories of decisions.

Anticipation (6 out of 20 incidents)

Nurses' experience with typical cases helped them to anticipate potential treatment requirements (anticipation is also described in the context of preparedness in the previous section). For instance, knowing that a 9-year old

patient with chest pain had been on medication for Attention Deficit Disorder (amphetamine/dextroamphetamine- which changes heart rhythm), the nurse anticipated that the patient would need cardiac monitoring and assigned the patient to a room with a monitor.

Awareness of risks and potential errors (6/20 incidents)

Nurses showed an awareness of various risks and potential misinformation based on their experience. A nurse, for instance, decided to move a patient who came in with a gunshot wound, to an inner room and erased his name from the whiteboard. She did this knowing from experience, that such patients have safety risks, and had taken similar precautions with such patients in the past. This a standard practice in many EDs that frequently care for patients with gunshot wounds. This signals to other staff members and clerical staff that if someone calls looking for the patient their name cannot be released because it may be media or potentially the shooter trying to locate the patient. Similarly, experience of ambiguity or misinformation from sensors made a nurse double-check an anomalous reading on a heart rate monitor with an EKG that provides a higher level of detail for interpretation. Awareness of risks and ambiguity of information primarily informed diagnosis and monitoring but was also associated with other decision categories such as preparedness.

Perceptions of Novice Nurses' Decision Making

In addition to themes identified in the incident narratives, participants also responded to the question *"What would a novice nurse do (in your place in this situation)?"*. Responses to this question helped identify the perspectives of participants on novice nurses, regardless of their own experience as a nurse. Prominent perspectives are described below:

Unfamiliarity/Lack of Knowledge (7/20 incidents)

Several participants thought that novice nurses would not be aware of nuanced procedures: *"Novice wouldn't have known to order lactic acid (for sepsis)"; "Novice wouldn't have known to hang two [liters of fluids]"*. This theme extends to the lack of awareness of alternate protocols or pathways: *"When you're trained, it's very systematic in that you have to do things in this order or this way. Experience teaches you to be flexible and prioritize on need basis"*. The lack of knowledge also extends to the availability of resources: *"If it were the nurses first time triaging, she may not have looked at the medication history, but also may not have connected Vyvanse and the class of drug (stimulants are cardiotoxic). New nurses get intimidated doing an EKG on a pediatric patient"*.

Lack of Situation Awareness (5/20 incidents)

Several participants mentioned that novice nurses tend to be more narrowly focused on their specific tasks and interactions with the patients and are less aware of some of the parallel processes and cues related to the patient. The ability to sense and heed multiple parallel cues is especially important for patients who may be having multiple co-morbidities or complications, or when the patient's condition is changing rapidly, all of which require close monitoring and sensing of multiple diagnostic indicators. Another implication of the narrow focus on the present is that they are less able to anticipate and plan ahead: "*Novice nurses usually think about all the present tasks that need to be done compared to thinking ahead and planning on future anticipated steps for smoother and more productive workflow"*.

Deference to Expertise (5/20 incidents)

Participants mentioned that novice nurses often tend to defer to more experienced colleagues in the ED for procedures they may not be familiar with, or decisions that they may not be comfortable taking: *"Less experienced nurse would have been more nervous, would have asked for assistance from a more experienced nurse"*.

DISCUSSION

Nurse participants described critical incidents or cases from recall, involving multiple decision points. Unlike the majority of previous literature on ED nurse decision making, which was centered on the triage process, this study focused on nurses' decision making about specific patients' critical situations. The decisions in these situations were characterized by the recognition of cues, situation assessment and problem identification, and nurse interventions. In particular, the role of experience was reflected in the way cues were identified and used, in their nuanced knowledge of clinical processes, and awareness of potential risks. These themes are broadly consistent with the nurse decision-making literature. For instance, nurse experience has been linked with clinical reasoning and critical thinking (Benner, 1984; Johansen & O'Brien, 2016; Simmons, 2010; Simmons et al., 2003). It is known that experience with specific patient populations or clinical situations allows nurses to recognize the clinical relevance of cues and distinguish between similar cues (Benner et al., 2011). We found further evidence of this, as in the incident in which the nurse realized that a patient's confusion seemed different than a regular migraine.

This study highlights the importance of intervention and adaptation in nurse decision making. In several instances, nurses' experience allowed them to choose or adapt a course of action that was specific to an atypical situation, particularly when the situation did not match a familiar or previously defined mental model. This is

also similar to findings by (Patel et al., 2008) who showed that experience led to guidelines being tailored to specific situations by nurses during the triage process. This particular characteristic of nurses' decision making is also similar to the notion of 'initiative' as defined by Woods (2019): "the ability of a unit to adapt when the plan no longer fits the situation, as seen from the unit's perspective". Adaptation is key to effective functioning under conditions of variability, which is characteristic of emergency medicine.

There were no clear differences in the description of incidents between nurses at the lower end of the experience scale (1.5 years) and upper end (>7 years). Regardless of experience, nurses tended to focus on similar cues and aspects in the situations described. This finding seems to contrast with previous literature which has shown systematic differences in themes between experienced and novice nurses (Benner, 1984). For instance, intuition, which has frequently been associated with experts' decision making, wasn't identified as a distinguishing factor in our study (Cone & Murray, 2002; Hicks et al., 2003). This may be, at least in part, because all our participants had sufficient expertise, i.e. had at least 1.5 years of experience in emergency nursing. In addition, since we made the methodological choice to give the participants broad latitude in selecting incidents to describe, they may have chosen incidents where they felt comfortable with their level of expertise. This methodological limitation may have biased our data towards examples in which the nurses had the most expertise to share, or to influence the decisions that were made in those situations.

In contrast, when participants were explicitly asked about "novice" behavior, their responses did reveal contrasts with expertise or their own level of experience. In particular, several participants' responses indicated the lack of factual knowledge about procedures and lack of situational awareness (SA) among novice nurses. SA in decision making involves the perception and processing of relevant information in the environment, assessment of the current situation, and projecting into the future based on knowledge of the environment and perceived cues (Endsley, 1995). This perception about novice nurses is consistent with the extant literature on nurses' decision making and expertise (Benner, 1984; Benner & Tanner, 1987; Cioffi, 1998). Several other themes identified in this category (perceptions about novice nurses) contrasted with characteristics of experienced nurses we identified through the incident descriptions under the 'role of experience' category: while novices deferred to expertise or to senior colleagues, more experienced nurses were able to take initiative under critical situations; experiential and factual knowledge of our participants contrasted with their perception that novice nurses lacked important knowledge that is gained through experience; similarly our participants provided multiple illustration of anticipation that drew on past experience, whereas they perceived novices as lacking in anticipation. This helps further clarify the role of experience in emergency nurses' decision making.

Limitations

Our primary limitation is that the data collection took place in 2017 therefore the study would not identify any challenges or themes related to current health IT systems; however since the analysis primarily focused on patient related assessments and decision-making, the conclusions would not be impacted. Moreover, current literature does not expand into critical decision making for nursing in the emergency department which highlights the relevance and novelty of this work. The least amount of experience of the nurses in our study was 1.5 years of ED experience. While this lower end of the range represents a relatively inexperienced category, the study did not involve nurses with less than a year, or less than 6 months of experience. Cowin et al (2006) found that it can take six months or longer for new nurses to adjust to their roles and develop confidence, before seeing an exponential increase in knowledge gained through experience. Therefore, including participants who are less than six months into nursing practice could allow for greater contrast between experience levels. Additionally, this was not a comparative study involving non-ED settings/nurses. Therefore, it is difficult to identify any of the themes in this paper as unique to ED nurses. That said, it is reasonable to highlight certain themes which are characteristic of the high-tempo, medically diverse context of the ED.

Implications for Future Research and Practice

Our study lays the ground for understanding nurse decision making in a broad emergency department context. The decision categories and characteristics identified, along with the role of experience, could form the basis for future research focusing on one or more of these aspects. For instance, future work could be done to find any systematic differences in decision making characteristics or strategies between various levels of experience in the ED. The decision categories reported here could be expanded upon in research focusing on the design of information displays and other decision aids for ED nurses. Finally, training programs for ED nurses could benefit from our findings in developing curricula around decision making.

CONCLUSION

This research demonstrated the novel use of CDM to better understand the nurse decision making during the ongoing care of patients in the emergency department. This methodology provided important insights into decision making, including the identification of important decision categories as well as the role of experience in decision making in emergency nursing. Further, the study identified nurses' perceptions of the role of experience

in decision making. The research also identified the role of nursing intervention and planning as critical decision making strategies to effectively care for patients in the dynamic environment of the Emergency Department. Future research is needed to better differentiate between experience levels and the development of tools and processes that better support emergency medicine nursing workflow.

ACKNOWLEDGEMENT

This work was support by grant number R01 HS022542 from the Agency for Healthcare Research and Quality.

REFERENCES

Agius, S., Magri, C., & Cassar, V. (2025). A Cognitive Task Analysis for Developing a Clinical Decision Support System for Emergency Triage. *Journal of Emergency Nursing*.

Aitken, L. M., Marshall, A. P., Elliott, R., & McKinley, S. (2009). Critical care nurses' decision making: sedation assessment and management in intensive care. *Journal of Clinical Nursing*, *18*(1), 36–45. http://onlinelibrary.wiley.com/doi/10.1111/j.1365-2702.2008.02318.x/abstract

Benner, P. (1984). *From novice to expert: Excellence and power in clinical nursing practice*. Addison-Wesley Publishing.

Benner, P., Kyriakidis, P. H., & Stannard, D. (2011). *Clinical wisdom and interventions in acute and critical care: A thinking-in-action approach* (2nd ed.). Springer Publishing Co.

Benner, P., & Tanner, C. (1987). Clinical judgement: How expert nurses use intuition. *American Journal of Nursing*, *87*(1), 23–31.

Chung, J. Y. M. (2005). An exploration of accident and emergency nurse experiences of triage decision making in Hong Kong. *Accident and Emergency Nursing*, *13*(4), 206–213. http://www.internationalemergencynursing.com/article/S0965-2302(05)00070-6/fulltext

Cioffi, J. (1998). Decision making by emergency nurses in triage assessments. *Accident and Emergency Nursing*, *6*(4), 184–191.

Cioffi, J. (2001). A study of the use of past experiences in clinical decision making in emergency situations. *International Journal of Nursing Studies*, *38*(5), 591–599. https://doi.org/10.1016/S0020-7489(00)00096-1

Cone, K. J., & Murray, R. (2002). Characteristics, insights, decision making, and preparation of ED triage nurses. *Journal of Emergency Nursing*, *28*(5), 401–406. http://www.jenonline.org/article/S0099-1767(02)00034-X/fulltext

Considine, J., Botti, M., & Thomas, S. (2007). Do Knowledge and Experience Have Specific Roles in Triage Decision-making? *Academic Emergency Medicine*, *14*(8), 722–726. http://onlinelibrary.wiley.com/store/10.1197/j.aem.2007.04.015/asset/j.aem.2007.04.015.pdf?v=1&t=jdhp09ee&s=4da916140c15603a2234e84b3918d5f36352e326

Cowin, L. S., & Hengstberger-Sims, C. (2006). New graduate nurse self-concept and retention: A longitudinal survey. *International Journal of Nursing Studies*, *43*(1), 59–70. https://doi.org/10.1016/J.IJNURSTU.2005.03.004

Crandall, B., & Getchell-Reiter, K. (1993). Critical decision method: A technique for eliciting concrete assessment indicators from the intuition of NICU nurses. *Advances in Nursing Science*, *16*(1). https://doi.org/10.1097/00012272-199309000-00006

Croskerry, P. (2005). The theory and practice of clinical decision-making. *Canadian Journal of Anesthesia*, *52*(6), R1–R8. https://doi.org/10.1007/BF03023077

Ebright, P. R., Patterson, E. S., Chalko, B. A., & Render, M. L. (2003). Understanding the Complexity of Registered Nurse Work in Acute Care Settings. *Journal of Nursing Adminstration*, *33*(12), 630–638.

Endsley, M. R. (1995). Toward a Theory of Situation Awareness in Dynamic Systems. *Human Factors: The Journal of the Human Factors and Ergonomics Society*, *37*(1). https://doi.org/10.1518/001872095779049543

Gerdtz, M. F., & Bucknall, T. K. (1999). Why we do the things we do: Applying clinical decision-making frameworks to triage practice. *Accident and Emergency Nursing*, *7*(1), 50–57. https://doi.org/10.1016/S0965-2302(99)80103-9

Harenčárová, H. (2015). Structured Analysis of Critical Decision Method Data – Emergency Medicine Case Study. *Human Affairs*, *25*(4), 443–459. https://doi.org/10.1515/HUMAFF-2015-0036

Hicks, F. D., Merritt, S. L., & Elstein, A. S. (2003). Critical Thinking and Clinical Decision Making In Critical Care Nursing: A Pilot Study. *Heart & Lung*, *32*(3), 169–180. https://doi.org/10.1016/S0147-9563(03)00038-4

Hobgood, C., Villani, J., & Quattlebaum, R. (2005). Impact of Emergency Department Volume on Registered Nurse Time at the Bedside. *Annals of Emergency Medicine*, *46*(6), 481–489. https://doi.org/10.1016/J.ANNEMERGMED.2005.07.014

Hoffman, K. A., Aitken, L. M., & Duffield, C. (2009). A comparison of novice and expert nurses' cue collection during clinical decision-making: verbal protocol analysis. *International Journal of Nursing Studies*, *46*(10), 1335–1344. http://www.journalofnursingstudies.com/article/S0020-7489(09)00126-6/fulltext

Hoffman, R., Crandall, B., & Shadbolt, N. (1998). Use of the critical decision method to elicit expert knowledge: A case study in the methodology of cognitive task analysis. *Human Factors*, *40*(2), 254–276.

Hoffman, R. R. (2005). *Protocols for Cognitive Task Analysis*. https://apps.dtic.mil/sti/citations/ADA475456

Hoffman, R. R., & Coffey, J. W. (2004). Weather Forecasting and the Principles of Complex Cognitive Systems. *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, *48*(3), 315–319. https://doi.org/10.1177/154193120404800309

Huang, J., Neill, L., Wittbrodt, M., Melnick, D., Klug, M., Thompson, M., ... & Etemadi, M. (2023). Generative artificial intelligence for chest radiograph interpretation in the emergency department. *JAMA network open*, *6*(10), e2336100-e2336100.

Jackson, J., Anderson, J. E., & Maben, J. (2021). What is nursing work? A meta-narrative review and integrated framework. *International journal of nursing studies*, *122*, 103944.

Johansen, M. L., & O'Brien, J. L. (2016). Decision Making in Nursing Practice: A Concept Analysis. *Nursing Forum*, *51*(1), 40–48. https://doi.org/10.1111/NUF.12119

Klein, G. A., Calderwood, R., & MacGregor, D. (1989). Critical decision method for eliciting knowledge. *IEEE Transactions on Systems, Man, and Cybernetics*, *19*(3), 462–472. https://doi.org/10.1109/21.31053

Laxmisan, A., Hakimzada, F., Sayan, O. R., Green, R. A., Zhang, J., & Patel, V. L. (2007). The multitasking clinician: decision-making and cognitive demand during and after team handoffs in emergency care. *International Journal of Medical Informatics*, *76*(11–12), 801–811. https://www.sciencedirect.com/science/article/pii/S1386505606002413?via%3Dihub

Mizerek, E., & Wolf, L. (2015). To Foley or Not To Foley: Emergency Nurses' Perceptions of Clinical Decision Making in the Use of Urinary Catheters in the Emergency Department. *Journal of Emergency Nursing*, *41*(4), 329–334. https://doi.org/10.1016/J.JEN.2014.09.009

Patel, V. L., Gutnik, L. A., Karlin, D. R., & Pusic, M. (2008). Calibrating urgency: triage decision-making in a pediatric emergency department. *Advances in Health Sciences Education*, *13*(4), 503–520. https://link.springer.com/article/10.1007%2Fs10459-007-9062-6

Patterson, C., Procter, N., & Toffoli, L. (2016). Situation awareness: when nurses decide to admit or not admit a person with mental illness as an involuntary patient. *Journal of Advanced Nursing*, *72*(9), 2042–2053. https://doi.org/10.1111/JAN.13024

Patterson, M. D., Militello, L. G., Bunger, A., Taylor, R. G., Wheeler, D. S., Klein, G., & Geis, G. L. (2016). Leveraging the Critical Decision Method to Develop Simulation-Based Training for Early Recognition of Sepsis. *Journal of Cognitive Engineering and Decision Making*, *10*(1), 36–56. https://doi.org/10.1177/1555343416629520

Potter, P., Boxerman, S., Wolf, L., Marshall, J., Grayson, D., Sledge, J., & Evanoff, B. (2004). Mapping the Nursing Process: A New Approach for Understanding the Work of Nursing. *JONA: The Journal of Nursing Administration*, *34*(2), 101–109.

Reay, G., Smith-MacDonald, L., Then, K. L., Hall, M., & Rankin, J. A. (2020). Triage emergency nurse decision-making: Incidental findings from a focus group study. *International Emergency Nursing*, *48*, 100791. https://doi.org/10.1016/J.IENJ.2019.100791

Schubert, C. C., Denmark, T. K., Crandall, B., Grome, A., & Pappas, J. (2013). Characterizing novice-expert differences in macrocognition: An exploratory study of cognitive work in the emergency department. *Annals of Emergency Medicine*. https://doi.org/10.1016/j.annemergmed.2012.08.034

Simmons, B. (2010). Clinical reasoning: concept analysis. *Journal of Advanced Nursing*, *66*(5), 1151–1158. https://doi.org/10.1111/J.1365-2648.2010.05262.X

Simmons, B., Lanuza, D., Fonteyn, M., Hicks, F., & Holm, K. (2003). Clinical Reasoning in Experienced Nurses. *Western Journal of Nursing Research*, *25*(6). https://doi.org/10.1177/0193945903253092

Ten Berg, H., van Bakel, B., van De Wouw, L., Jie, K. E., Schipper, A., Jansen, H., ... & Kurstjens, S. (2024). ChatGPT and generating a differential diagnosis early in an emergency department presentation. *Annals of emergency medicine*, *83*(1), 83-86.

Wong, B. L. W. (2004). Critical decision method data analysis. In D. Diaper & N. A. Stanton (Eds.), *The handbook of task analysis for human-computer interaction* (pp. 327–346). Lawrence Erlbaum Associates, Inc.

Woods, D. D. (2019). Essentials of resilience, revisited. In M. Ruth & S. Goessling-Reisemann (Eds.), *Handbook on Resilience of Socio-Technical Systems* (pp. 52–65). Edward Elgar Publishing Ltd. https://doi.org/10.4337/9781786439376.00009